Quick Start for Supercomputers (Version 2026.1)

0. Introductions for clusters of CCSE

There are two hybrid clusters managed by CCSE (Center for Computational Science and Engineering), which provide computational services to the PI groups of SUSTech. They are established in 2015 and 2018 respectively. The two clusters are called 'TaiYi' and 'QiMing'. 'QiMing' was updated and expanded in 2023, so we now call it 'QiMing v2.0'.

1. Login the clusters

a) Login to 'QiMing v2.0': *ssh -p 18188 username@172.18.6.10 ~ 12*

b) Login to 'TaiYi': *ssh username@172.18.6.175.* If you have graphical requirements, please *ssh username@172.18.6.176.*

2. Queue configurations of the clusters

a) 'QiMing v2.0':

Queue	Cores	Mem.	CPU Type	GPU Type	*Run Limit (h)	Nodes	Remarks
38	64	512g	3rd Gen Xeon Sca.	-	720	36	CPU queues
33	40	192g	1st Gen Xeon Sca.	-	720	108	
ot38	64	2t**	3rd Gen Xeon Sca.	-	336	1	
2t50c	64	2t	3rd Gen Xeon Sca.	-	336	4	
1t75c	64	1t	3rd Gen Xeon Sca.	-	336	10	
1t88c	96	1t	4th Gen Xeon Sca.	-	336	20	
1t81c	120	1t	5th Gen Xeon Sca.	-	336	30	
73x	128	512g	3rd Gen EPYC	-	336	2	
52	48	512g	2nd Gen EPYC	-	336	1	
63	128	512g	3rd Gen EPYC	-	336	1	
v4-3t	72	3t	Xeon E5 v4	-	336	1	
sk-6t-104	104	6t	1st Gen Xeon Sca.	-	336	1	
75m	48	192g	1st Gen Xeon Sca.	-	720	2	
v3-6t	144	6t	Xeon E7 v3	-	336	7	
v3-64	24	64g	Xeon E5 v3	-	1440	225	
spec-cpu	-	-	-	-	Unlimited	-	
2h100-80	192	768g	4th Gen EPYC	2*h100 80g	336	1	GPU queues
2a100-40	40	192g	2nd Gen Xeon Sca.	2*a100 40g	336	1	
4a100-40	56/64	384/512g	2nd Gen Xeon Sca.	4*a100 40g	336	3	
2a100-80	40	192g	2nd Gen Xeon Sca.	2*a100 80g	336	1	
4a100-80	160/128/96/32	128g/512g	4th/3rd/2nd Gen EPYC	4*a100 80g	336	6	
hgx	64/128	1t	3rd EPYC/Xeon Sca	8*a100 80g	336	5	
8v100-32-sc	48	384g	2nd Gen Xeon Sca.	8*v100 32g	336	2	
4v100-16-e5	24	64g/128g	Xeon E5 v3	4*v100 16g	336	2	
4v100-16-sc	40	192g	1st Gen Xeon Sca.	4*v100 16g	336	1	
2v100-16-sc	40	384g	1st Gen Xeon Sca.	2*v100 16g	336	3	
2v100-32-e5	24	256g	Xeon E5 v3	2*v100 32g	336	1	
2v100-32-sc	40	192g	2nd Gen Xeon Sca.	2*v100 32g	336	1	
spec-gpu	-	-	-	-	Unlimited	-	

b) 'TaiYi':

Queue	Cores	Mem.	CPU Type	GPU Type	*Run Limit (h)	Nodes	Core limit for single job
large	40	192g	1st Gen Xeon Sca.	-	720.00	811	More than 600 cores
medium	40	192g	1st Gen Xeon Sca.	-	720.00		More than 200 cores
short	40	192g	1st Gen Xeon Sca.	-	720.00		More than 40 cores
ser	40	192g	1st Gen Xeon Sca.	-	168.00		Less than 100 cores
debug	40	192g	1st Gen Xeon Sca.	-	0.33		Less than 80 cores
gpu	40	384g	1st Gen Xeon Sca.	2*v100 16g	168.00	1.00	20 cores / gpu card
spec	40	192g	1st Gen Xeon Sca.	-	Unlimited	-	-
smp	192	6t	1st Gen Xeon Sca.	-	168.00	2.00	-

**The node's memory in queue ot38 is 512 GB DDR4 and 2 TB Optane.

*The RUNLIMIT time of the queues will be changed depending on the usage of the queue. You can get it with the following command:

"*bqueues -l* {the queue name} | *grep RUNLIMIT -A 1*"

#The number of nodes in each queue will change as the cluster expands. You can get it with the following command:

"*bhosts hg_{the queue name} | wc -l | awk '{print $1-1}'*"

3. Job Script Examples

Both the 'QiMing v2.0' and 'TaiYi' job examples are stored in the directory of /share/examples/. The *exam1* and *exam2* are examples of CPU jobs running on single node and multi-node, respectively. The *exam3* and *exam4* are examples of GPU jobs running on single node and multi-node, respectively.

4. Frequently Used Commands of LSF

Commands	Description	Frequently used methods
bsub	Submits a job to LSF	*bsub < jobs.lsf*
bjobs	Displays the LSF jobs.	*bjobs; bjobs -l; bjobs -l JOBID*
bkill	Sends signals to kill the LSF jobs	*bkill JOBIDs*
bhosts	Displays hosts and their status	*bhosts; bhosts -w -gpu*
bqueues	Displays queues' information	*bqueues; bqueues -l QUEUENAME*
lsload	Displays hosts' loading information	*lsload; lsload -gpuload*

5. The Users' Storage Directories

a) 'QiMing v2.0':

i. /scratch: It is used to store the temporary data of your running jobs for free, and the temporary data can only be stored for 8 days before automatically cleaning. Please back up your data in time.

ii. /work: It is used to store your job results and software. Each group has 1, 3 or 5 TB of free capacity depending on the type of account.

b) 'TaiYi':

i. /work: It is used to store your software. Each group has 1 TB of free capacity.

ii. /data: It is used to store your job results. Each group has 2 or 4 TB of free capacity depending on the type of account.

iii. /scratch: It is used to store the temporary data of your running jobs for free, and the temporary data can only be stored for 10 days before automatically cleaning. Please back up your data in time.

You can view your group's or user's storage quota with the following command: *mmlsquota -g(u) groupname(username) --block-size auto.*

6. Fees:

Fees: you can view the standard fees for each queues and storages by visiting https://hpc.sustech.edu.cn/business/chargingstandard.html.

7. Some Tips

a) Please submit your jobs by LSF system, and don't run your jobs on login nodes for long time.

b) After submitting your jobs, please check that the jobs are running correctly.

c) If your jobs have been pending for a long time, please check whether your resource applications in LSF are correct (#BSUB -R "…"; #BSUB -n "…"; #BSUB -gpu "…").

d) The full user guide is available at https://hpc.sustech.edu.cn.

8. Suggestions and Seeking helps

If you have any suggestions or problems with the use of the supercomputers, please do not hesitate to submit them via the Issue Tracking System (ITS).
The ITS's web site is: https://ums.hpc.sustech.edu.cn/.
The ITS's tutorial can be found at:
https://hpc.sustech.edu.cn/ref/HPMS_UserGuide.pdf.

Customer Service Tel.: 0755-88015833.